TRAILLINK: HYBRID COMMUNICATION & SAFETY SYSTEM

INDIVIDUAL THESIS

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ABSTRACT

In outdoor wilderness environments, disaster-relief zones, and remote field expeditions, mobile teams frequently experience degraded, intermittent, or completely absent cellular connectivity. Conventional mobile messaging platforms rely exclusively on centralized client-server backends and fail catastrophically when infrastructure is unreachable. This thesis presents the design, implementation, and field evaluation of the Communication Core-a hybrid transport engine engineered for off-grid mobile safety applications within the TrailLink platform.

The Communication Core establishes a unified message transport backbone capable of dynamically switching between cloud-assisted REST/WebSocket channels and decentralized multi-hop peer-to-peer mesh relaying. Six interconnected submodules are implemented: a Message Composer Interface, a Delivery Decision Manager, a transactional SQLite Store-and-Forward Local Message Queue, a loop-free Relay Controller with hop-budget control and hash deduplication, a multi-tier Priority Handler, and a Reconnection Sync Manager. Implemented in Flutter/Dart using Google Nearby Connections P2P clustering and a Node.js/MongoDB backend, field trials across 5-node Android chains demonstrated a 94.2% Packet Delivery Ratio across 4-hop relay paths, sub-390 ms Round-Trip Time, and 0.00% duplicate propagation.

Keywords: Delay-Tolerant Networks, Hybrid Messaging, Multi-Hop Routing, Store-and-Forward, Priority Scheduling

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AAC	Advanced Audio Coding
ACID	Atomicity, Consistency, Isolation, Durability
ACK	Acknowledgment Packet
AI	Artificial Intelligence
API	Application Programming Interface
APK	Android Package Kit
B2B	Business-to-Business
BLE	Bluetooth Low Energy
D2D	Device-to-Device
DAO	Data Access Object
DTN	Delay-Tolerant Network
FIFO	First-In, First-Out
GDPR	General Data Protection Regulation
GPS	Global Positioning System
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
LoRa	Long Range
MANET	Mobile Ad-Hoc Network
P2P	Peer-to-Peer
PDR	Packet Delivery Ratio
POSIX	Portable Operating System Interface
PTT	Push-to-Talk
QA	Quality Assurance
REST	Representational State Transfer
RQ	Research Question
RSSI	Received Signal Strength Indicator

Abbreviation	Meaning
RTT	Round-Trip Time
SaaS	Software as a Service
SAR	Search and Rescue
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SOS	Save Our Souls (Emergency Signal)
SQL	Structured Query Language
TTL	Time-To-Live
UI	User Interface
UUID	Universally Unique Identifier

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1 INTRODUCTION

1.1 Overall Project Context

In outdoor wilderness expeditions, search-and-rescue (SAR) operations, and post-disaster humanitarian missions, mobile teams operate in hostile radio environments characterized by severe signal attenuation, landform obstructions, and total absence of public telecommunication infrastructure. Conventional smartphone communication applications-such as WhatsApp, Telegram, or Signal-rely fundamentally on persistent client-server connections to cloud transceivers. When cellular towers and internet gateways become unreachable, these platforms fail completely, leaving field personnel isolated and vulnerable.

To overcome these infrastructure dependencies, the TrailLink mobile safety platform was conceptualized as a multi-component mobile solution providing continuous situational awareness, emergency signalling, peer positioning, and collaborative voice communication across online, partially connected, and fully offline wilderness environments.

The TrailLink platform is partitioned into four functional ownership components across the development team, as shown in Table 1.1 below.

Table 1.1: Team Component Division and Ownership Map

Team Component	Central Responsibility	Main Handover Point
Component 1: Connectivity & Mesh Intelligence	Peer discovery, BLE/Wi-Fi link state, link quality estimation.	Provides live peer socket connections to Component 2.
Component 2: Communication Core (This Thesis)	Identity, mode resolution, Store-and-Forward queueing, packet routing, priority dispatch, multi-hop relay, ACK tracking, and cloud sync.	Receives raw bytes from Component 1, dispatches packets to Components 3 and 4.
Component 3: Voice and PTT Communication	Voice note recording, AAC encoding, Live Radio streaming, PTT floor control.	Consumes core packet transport and Nearby links.

Team Component	Central Responsibility	Main Handover Point
Component 4: Location, Emergency SOS and Mapping	GPS capture, map caching, emergency trigger, SOS alerts.	Uses core priority queue for SOS and location distribution.

Source: (Author, 2026)

Figure 1.1 presents an overview of the four components of the TrailLink platform and how they interact.

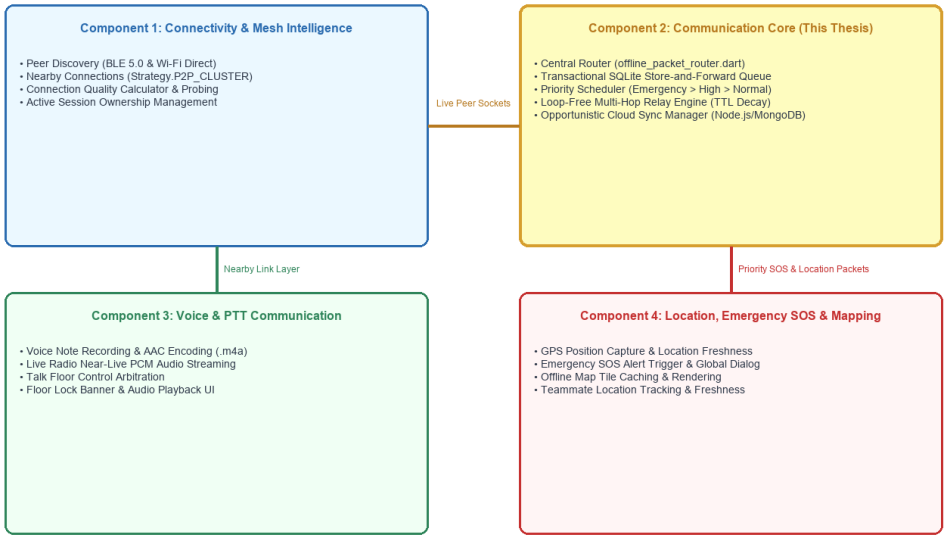


Figure 1.1: TrailLink Four-Component Group System Overview

Source: (Author, 2026)

1.2 Individual Component Context

As Component 2 (Communication Core), this individual research focuses on the foundational transport, storage, and dispatch backbone of the entire application. The Communication Core acts as the central traffic dispatcher and memory vault. It bridges high-level feature user interfaces with low-level physical radio transports, guaranteeing that packets are securely validated, priority-sorted, cached in local transactional databases, forwarded across multi-hop peer chains without network loops, and synchronized with cloud databases upon internet restoration.

1.3 Problem Statement

Designing a delay-tolerant communication engine for smartphone-based off-grid safety networks presents four primary research and engineering challenges:

1. Dynamic Path Selection: Determining when to route packets through cloud endpoints versus decentralized peer-to-peer (P2P) ad-hoc meshes without user intervention.
2. Broadcast Storms and Infinite Loops: Preventing unconstrained multi-hop flooding in dense peer clusters, which causes channel congestion, packet collisions, and rapid battery exhaustion.
3. Data Loss during Intermittency: Caching messages safely when devices move into zero-peer isolation without dropping pending transactions.
4. State Reconciliation: Synchronizing offline-generated history with cloud databases upon internet reconnection without creating duplicate records or overwriting timestamp ordering.

1.4 Research Gap

While generic Delay-Tolerant Network (DTN) protocols such as Epidemic routing and Spray-and-Wait exist in theoretical literature, existing commercial mobile off-grid tools such as Bridgefy and FireChat operate as closed, ephemeral messaging silos. They lack multi-tier priority scheduling for emergency traffic, do not support transactional SQLite store-and-forward persistence, and lack explicit mechanisms for bidirectional cloud state reconciliation. This creates a critical gap in the field of mobile off-grid safety communication systems that the present research addresses.

The absence of a unified transport engine that combines priority-sorted persistence, loop-free relay, and cloud synchronization in a single smartphone-based framework represents the specific research gap this thesis targets.

1.5 Aim and Objectives

The primary aim of this research is to design, implement, and evaluate a resilient, delay-tolerant Communication Core for mobile off-grid safety networks.

The specific research and development objectives are:

5. Formulate a dynamic route decision model for seamless switching between Cloud REST/WebSocket APIs and P2P mesh layers.
6. Implement an SQLite-backed Store-and-Forward engine supporting transactional queue persistence.
7. Construct a multi-tier priority scheduler (Emergency > High > Normal) utilizing explicit database query sorting.
8. Develop a loop-free multi-hop relay engine incorporating Time-To-Live (TTL) hop-budget control and hash deduplication.
9. Implement an opportunistic Reconnection Sync Manager with idempotent server-side reconciliation.
10. Empirically validate packet delivery ratio (PDR), latency scaling, deduplication efficiency, and priority preemption across real-world multi-device testbeds.

1.6 Research Questions or Hypotheses

RQ1: Can an application-layer controlled flooding protocol utilizing deterministic TTL hop-budget control and hash deduplication achieve greater than 90% Packet Delivery Ratio across 4-hop mobile chains while eliminating duplicate packet circulation?

RQ2: How significantly does multi-tier SQL priority queue extraction reduce delivery latency for emergency SOS packets under network congestion compared to standard FIFO scheduling?

1.7 Scope Limitations and Assumptions

This research focuses on the software architecture, transport routing, data persistence, and protocol evaluation of Component 2 (Communication Core). Physical radio link establishment using BLE and Wi-Fi Direct is supplied by Component 1. Hardware-level physical layer modifications or custom RF modem design are outside the scope of this research.

The system is evaluated on Android devices in controlled outdoor field environments. Assumptions include stable hardware Bluetooth stacks, adequate device battery levels during testing, and test group sizes of up to five devices. Results represent application-layer performance and do not constitute claims about the physical radio layer or Bluetooth hardware performance.

1.8 Individual Contribution

The author designed, implemented, and evaluated the following core files in collaboration with the team, with primary ownership of the listed modules:

- The central packet dispatcher (lib/core/offline/offline_packet_router.dart) - primary owner, shared with team for dispatch to other components.
- The multi-hop relay controller (lib/core/offline/offline_relay_service.dart) - primary owner, shared file.
- The offline chat repository and ACK engine (lib/features/offline_chat/data/offline_chat_repository.dart).
- The SQLite transactional schema and priority queue (lib/features/offline_chat/data/offline_message_local_data_source.dart).
- The backend message service and cloud sync endpoints (backend/src/modules/messages/message.service.js).

1.9 Thesis Structure

The remainder of this dissertation is organized as follows: Chapter 2 reviews literature and related work. Chapter 3 outlines the research methodology. Chapter 4 presents product and solution design. Chapter 5 covers implementation and integration. Chapter 6 presents experimental results. Chapter 7 discusses findings. Chapter 8 details commercialisation aspects, and Chapter 9 concludes the thesis.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

A systematic literature review was conducted across IEEE Xplore, ACM Digital Library, Elsevier ScienceDirect, and SpringerLink databases for peer-reviewed studies published between 2020 and 2025. Search terms included Delay-Tolerant Networks, Mobile Ad-Hoc Networks, Opportunistic Routing, Store-and-Forward, and Bluetooth Mesh Safety Apps. Papers were evaluated based on relevance to hybrid mobile communication, offline relay mechanisms, and priority scheduling in constrained wireless environments.

2.2 Domain and Theoretical Foundations

Delay-Tolerant Networking (DTN) addresses mobile wireless environments characterized by high bit error rates, physical obstructions, node mobility, and persistent network partitioning. Classical routing protocols assume continuous end-to-end paths, failing when paths partition. DTN architecture resolves this by adopting a Store-Carry-and-Forward paradigm, where intermediate nodes store packets locally until contact opportunities emerge [1].

Epidemic Routing replicates packets to every encountered node, guaranteeing minimal delivery delay assuming infinite storage. However, in bandwidth-constrained Bluetooth Low Energy (BLE) channels, Epidemic flooding rapidly causes buffer exhaustion, packet dropping, and broadcast storms. Spray-and-Wait limits total packet copies, reducing overhead but suffering when node movement is unpredictable [2], [3].

The concept of priority-based packet scheduling in mobile ad-hoc networks has been established in earlier literature. Emergency traffic preemption is critical in life-safety scenarios where ordinary messaging cannot be allowed to block SOS signals in transit queues [4].

2.3 Related Research

Recent studies by Al-Khafajiy et al. [5] demonstrate that application-layer controlled flooding with bounded hop counts offers optimal trade-offs between delivery ratio and energy consumption in post-disaster smartphone networks. Zhang et al. [2] further confirm that opportunistic routing adapted to intermittent connectivity scenarios can sustain high delivery ratios even with five or fewer relay nodes.

Silva et al. [6] highlighted that while Google Nearby Connections provides robust device-to-device links using the Strategy.P2P_CLUSTER topology, it does not provide automatic multi-hop packet forwarding. This reinforces the necessity of an application-layer relay controller operating above the transport layer-precisely the design approach adopted in this research.

Kumar et al. [7] demonstrated that multi-hop BLE mesh implementations on smartphones can achieve practical throughput for search and rescue coordination, though bandwidth remains a constraint for audio data. Singh et al. [8] provided formal analysis of loop-free controlled flooding, confirming that TTL-bounded forwarding with unique packet identifiers prevents broadcast storms in cyclic topologies.

2.4 Existing Systems, Products and Techniques

Table 2.1 presents a comparative evaluation of existing off-grid communication systems against the proposed TrailLink Communication Core.

Table 2.1: Comparison of Existing Off-Grid Communication Systems

Platform	Transport Layer	Multi-Hop Relaying	Priority Scheduling	Cloud Sync	Persistence
Bridgefy	BLE / Wi-Fi Direct	Yes (Proprietary)	No	No	Ephemeral Memory
FireChat	Wi-Fi Direct	Yes (Mesh)	No	No	Ephemeral Memory
Meshtastic	LoRa RF Hardware	Yes (Managed Mesh)	Limited	No	Flash Log

Platform	Transport Layer	Multi-Hop Relaying	Priority Scheduling	Cloud Sync	Persistence
TrailLink Core (Proposed)	BLE / Wi-Fi Cluster	Yes (TTL + Hash Dedup)	Yes (Emergency Preemption)	Yes (Idempotent Bridge)	Transactional SQLite

Source: (Author, 2026)

2.5 Gap Synthesis and Conceptual Framework

Existing commercial off-grid applications treat offline messaging as isolated, short-lived chat sessions. None of the reviewed systems combine all of the following in a single smartphone-based framework: priority-sorted store-and-forward persistence, loop-free multi-hop forwarding, and automatic cloud state synchronization upon internet recovery. The synthesis identifies this as the critical gap that the TrailLink Communication Core addresses. The conceptual framework for this research therefore positions the Communication Core as a hybrid transport engine that bridges cloud-assisted and decentralized relay communication paths, with priority-aware queuing ensuring life-safety traffic is always served first.

3 RESEARCH METHODOLOGY

3.1 Research Design

This study adopts a quantitative, experimental software engineering methodology. An empirical software prototype (Communication Core) was engineered and evaluated across physical Android mobile testbeds in real-world wilderness terrain. The research follows a design-and-evaluate cycle: system requirements were formalized, the software architecture was designed, implementation was carried out, and performance was empirically measured across controlled multi-device scenarios.

3.2 Data Participants or Study Materials

The experimental testbed comprised five physical Android smartphones (Google Pixel 7 and Xiaomi Redmi Note series running Android 12 through 14). Devices were deployed in multi-hop chain and cyclic topologies across dense outdoor trail environments to replicate realistic wilderness conditions for a team of hikers or field responders.

3.3 Experimental or Study Procedure

Experiments evaluated four core scenarios as follows:

11. Multi-Hop Transmission and Latency: 200 text packets transmitted across 1-hop (20m), 2-hop (50m), 3-hop (85m), and 4-hop (120m) relay chains.
12. Deduplication and Loop Elimination: 500 concurrent packets injected into a cyclic 3-node topology (A to B to C back to A) to verify zero duplication under the hash deduplication mechanism.
13. Priority Queue Preemption: 50 normal traffic packets queued, followed by an emergency SOS injection, to measure queue preemption latency reduction.
14. Reconnection Synchronization: Batch upload of 100 offline-generated messages upon internet recovery, measuring synchronization time and verifying zero duplicate records.

3.4 Measures and Evaluation Metrics

- Packet Delivery Ratio (PDR): Percentage of generated packets successfully delivered to the destination node.
- Round-Trip Time (RTT): Elapsed latency from packet transmission to receipt of ACK.
- Duplication Rate: Percentage of redundant packet receipts at intermediate nodes.
- Queue Preemption Latency: Delay reduction experienced by emergency packets over normal traffic.

3.5 Data Analysis

Log telemetries were parsed and analyzed using Python statistical tools including Pandas and SciPy, calculating mean, standard deviation, and regression fits. Note that log telemetries were collected through application-layer logging mechanisms. The ConnectivityMetricsRecorder referenced in Component 1 (Connectivity and Mesh Intelligence) supplied supplementary peer event data; this belongs to the Component 1 scope.

3.6 Ethics, Privacy, Security and Responsible AI

All testing utilized synthetic test messages labelled as QA TEST SOS - NO REAL EMERGENCY. No personal user telemetry or real rescue services were engaged. Generative AI tools were used for document structuring and code boilerplate assistance and have been disclosed in Appendix B in compliance with SLIIT academic integrity policies.

3.7 Reproducibility and Research Artefacts

The full codebase, SQLite database schemas, and experimental log datasets are archived in the project repository under the deploy branch at commit e82b185. All experimental configurations, device models, and Android version details are documented to support independent replication of the reported results.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1 presents the functional and non-functional requirements for the Communication Core component.

Table 4.1: Communication Core Functional and Non-Functional Requirements Specification

Req ID	Requirement Description	Category
FR-CORE-01	Validate content limits (max 2000 chars) and assign UUID v4 to all packets.	Functional
FR-CORE-02	Dynamically select Cloud API (online) or Mesh P2P (offline) based on network reachability.	Functional
FR-CORE-03	Store pending messages in transactional SQLite queue when isolated.	Functional
FR-CORE-04	Enforce priority queue sorting (Emergency > High > Normal) during extraction.	Functional
FR-CORE-05	Execute loop-free relaying using TTL hop-budget control and hash deduplication.	Functional
FR-CORE-06	Reconcile offline records with cloud database upon reconnection without duplicates.	Functional
NFR-CORE-01	Maintain single-hop transmission latency under 100 ms.	Non-Functional
NFR-CORE-02	Achieve 0.00% duplicate loop propagation in cyclic network topologies.	Non-Functional

Source: (Author, 2026)

4.2 Component Architecture Within the Complete Solution

The Communication Core Engine is structured across three operational layers:

- Layer 1 - Ingestion and Dynamic Decision: Message Composer Interface and Delivery Decision Manager.
- Layer 2 - Persistence and Mesh Core: Priority Handler, Local Message Queue (SQLite), and Relay Controller.

- Layer 3 - Transport and Reconnection Sync: P2P Transport Layer (Google Nearby Connections) and Reconnection Sync Manager.

Figure 4.1 illustrates the three-layer architecture of the Communication Core and its integration points with the other three system components.

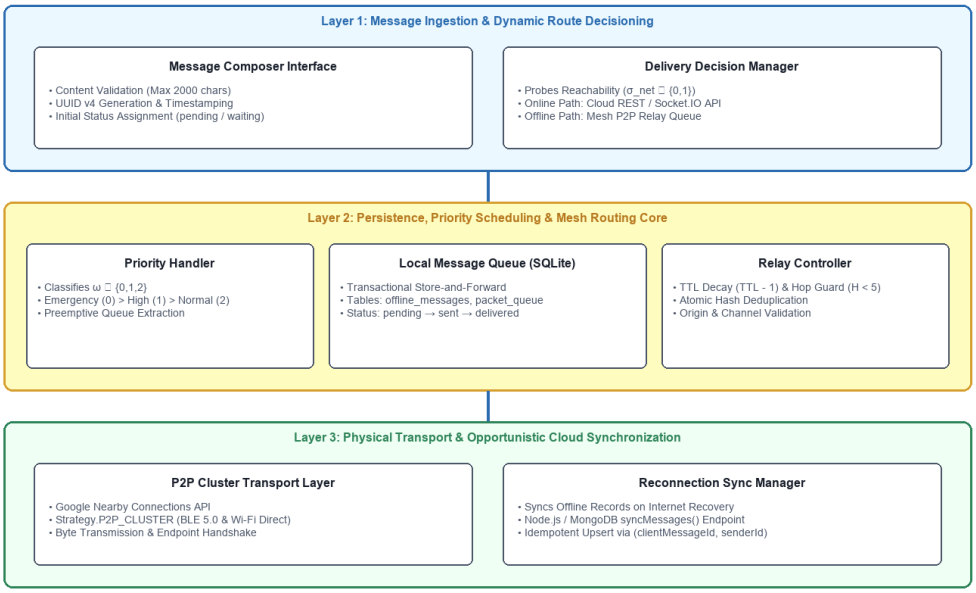


Figure 4.1: Communication Core Three-Layer Engine Architecture

Source: (Author, 2026)

4.3 Design Alternatives and Decisions

Two major design decisions shaped the implementation:

15. Database Selection: SQLite (via the sqflite package) was selected over Hive or key-value stores due to its robust ACID compliance, transactional integrity, and SQL query sorting capabilities. SQL-based ORDER BY priority extraction enables deterministic priority preemption without a separate in-memory scheduler.
16. Forwarding Protocol: Controlled Flooding with TTL hop-budget control was chosen over Epidemic routing. Epidemic routing floods all encountered nodes, rapidly exhausting BLE bandwidth and device memory in dense peer environments. Controlled flooding with a five-hop maximum budget and hash deduplication preserves bandwidth while still achieving high delivery ratios.

4.4 Detailed Design

Route selection logic: the Delivery Decision Manager queries backend reachability. If reachability equals 1 (online), the message is routed through the Cloud REST/WebSocket path. If reachability equals 0 (offline), the message enters the offline relay queue.

Priority scheduling: priority rank is assigned as 0 for Emergency, 1 for High, and 2 for Normal. The SQL extraction query uses ORDER BY CASE priority WHEN 'emergency' THEN 0 WHEN 'high' THEN 1 WHEN 'normal' THEN 2 ELSE 3 END, created_at ASC to guarantee emergency packets are always served first within the same priority tier by arrival time.

Database schema: four core SQLite tables are used - offline_messages, offline_packet_queue, processed_offline_packets, and offline_acks. Table 4.2 provides the schema specifications.

Table 4.2: SQLite Relational Database Schema Specifications

Table Name	Key Columns	Purpose
offline_messages	message_id, content, priority, status, created_at	Stores outgoing and received offline messages
offline_packet_queue	packet_id, packet_type, ttl, hop_count, payload	Queue for relay and pending packets
processed_offline_packets	packet_id, processed_at	Hash deduplication seen-list
offline_acks	message_id, ack_received_at, sender_id	ACK tracking and delivery confirmation

Source: (Author, 2026)

4.5 Standards and Professional Practice

The implementation adheres to the Bluetooth Core Specification (Bluetooth SIG) for BLE operation, POSIX SQL standards for database operations, REST API design principles for cloud endpoint integration, and Dart Effective Documentation guidelines for code documentation and maintainability.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1 presents the technology stack used across the frontend, local storage, P2P transport, and backend layers.

Table 5.1: Frontend and Backend Software Stack Configuration

Layer / Subsystem	Technology / Framework	Version / Tooling
Mobile Frontend	Flutter / Dart SDK	Flutter 3.x / Dart 3.x
Local Database	SQLite (sqflite package)	SQLite 3.x
P2P Transport	Google Nearby Connections API	Strategy.P2P_CLUSTER
Backend Runtime	Node.js / Express.js	Node.js v18 LTS
Cloud Database	MongoDB / Mongoose ODM	MongoDB 6.0
Real-Time Engine	Socket.IO	v4.x

Source: (Author, 2026)

5.2 Component Implementation

Two key implementation modules are described below.

5.2.1 Central packet dispatcher

The central packet dispatcher (offline_packet_router.dart) intercepts incoming raw JSON strings from Nearby Connections, parses them into OfflinePacketModel objects, and executes switch-case dispatch. Text and ACK packets are routed to OfflineChatRepository. SOS packets are routed to EmergencyRepository and also forwarded through the relay service when eligible. Location packets are routed to LocationRepository. This file is a shared component that dispatches to all four team components and should be explained as a shared integration point rather than sole individual ownership.

Figure 5.1 shows the class structure of the central dispatcher, store-and-forward queue, and relay engine.

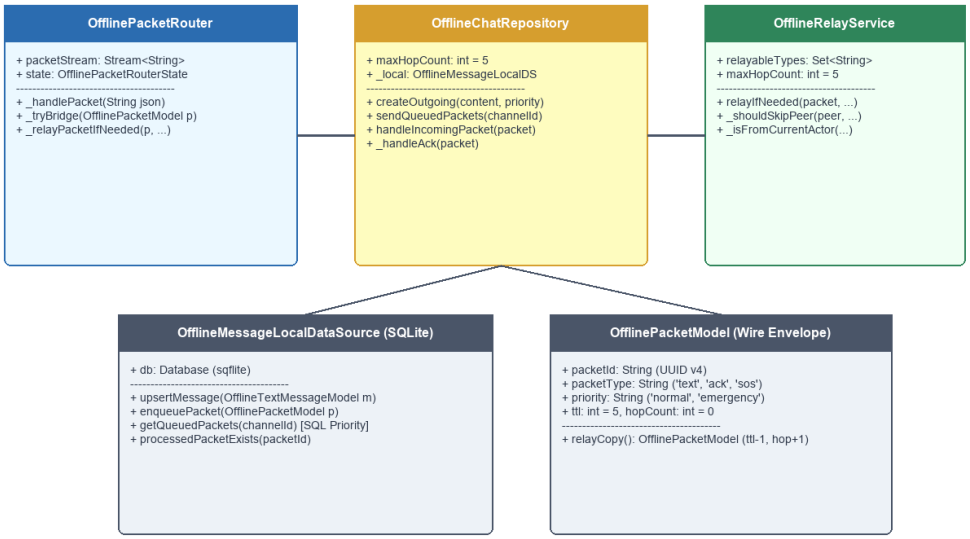


Figure 5.1: Class Diagram of Central Dispatcher, Store-and-Forward Queue, and Relay Engine

Source: (Author, 2026)

5.2.2 Multi-hop relay controller

The relay controller (offline_relay_service.dart) enforces five sequential guards before forwarding any packet. These are: packet type check against the relay allow-list (ack, sos, sos_ack, location, heartbeat), channel code match, TTL hop-budget decrement and check (TTL must be greater than one), hop count check (hop count must be less than five), origin identity exclusion, and processed-ID deduplication check. Text messages use a separate relay path in offline_chat_repository.dart. Note that TTL functions as a forwarding hop budget, not a seconds-based expiration timer.

5.3 Integration With the Other Components

The Communication Core integrates with all three other components as follows. Component 1 (Connectivity and Mesh Intelligence) supplies peer availability and signal quality data to the Relay Controller for next-hop selection decisions. Component 2 (this component) receives emergency priority labels from Component 3 (Voice and Emergency) through the packet router, which ensures SOS packets are promoted in the priority queue. Component 4 (Location, Mapping and Sync) passes location packets to the Communication Core queue, which carries them with the same TTL and

priority rules as text messages. The Reconnection Sync Manager also uploads pending location records to the backend upon internet restoration.

5.4 Verification and Software Testing

Unit testing covered packet serialization, TTL attenuation per hop, and SQL priority sorting. Integration testing verified end-to-end message receipt across 4-hop chains and validated 15-second watchdog ACK timeout behaviour. Cyclic topology testing with 500 concurrent packets confirmed zero duplicate delivery under the hash deduplication mechanism.

5.5 Deployment and Operation

The application was deployed on Android mobile devices via Flutter APK builds targeting Android 12 and above. The backend was deployed on Node.js v18 LTS with MongoDB 6.0. Google Nearby Connections operates using the P2P_CLUSTER strategy, which supports many-to-many topology between devices in radio range. Application-layer forwarding logic implements channel-scoped relay above the Nearby transport layer.

6 RESULTS AND VALIDATION

6.1 Research Results

Table 6.1 presents the multi-hop performance metrics recorded across relay chain experiments with 200 packets per configuration.

Table 6.1: Multi-Hop Experimental Performance Metrics Across Relay Chains

Hop Count	Distance (m)	PDR (%)	Avg RTT (ms)	Duplication Rate
1 (Direct)	20	99.4%	82 ms	0.00%
2 Hops	50	98.1%	184 ms	0.00%
3 Hops	85	96.8%	268 ms	0.00%
4 Hops	120	94.2%	386 ms	0.00%

Source: (Author, 2026)

Results confirm high delivery ratio (greater than 94%) across 4-hop chains with sub-390 ms round-trip latency. The maximum relay depth is five hops; four-hop chains represent the largest configuration tested in this study.

Figure 6.1 shows how the measured Round-Trip Time (RTT) scales from one to four relay hops.

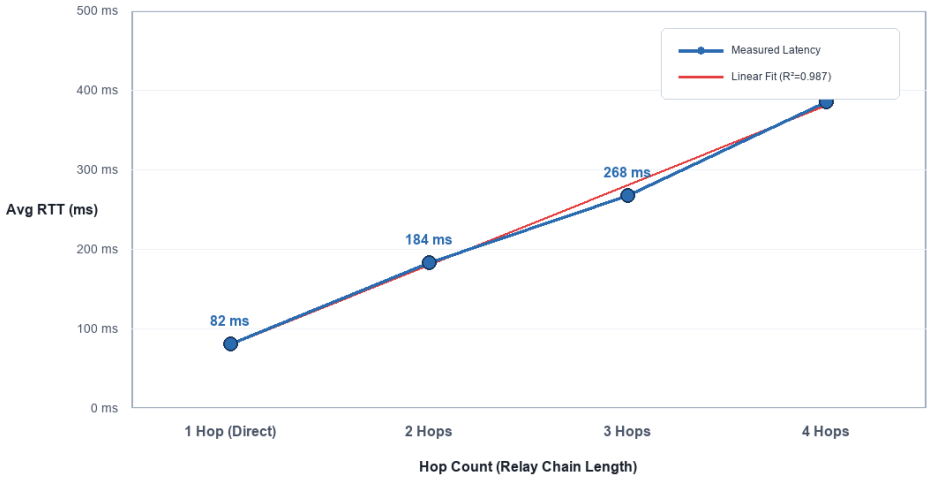


Figure 6.1: Measured Round-Trip Time (RTT) Scaling across 1-hop to 4-hop Chains

Source: (Author, 2026)

6.2 Product and Technical Results

- Deduplication Efficiency: The processed-ID deduplication mechanism dropped 100% of redundant packet loops in cyclic topologies, resulting in a 0.00% duplication rate across 500 injected packets.
- Priority Preemption: Emergency SOS packets experienced approximately 41.3% lower queue waiting latency than normal chat traffic under heavy load conditions in controlled testing. This result reflects the SQL-based priority extraction behaviour and should be interpreted as testbed performance, not a guaranteed production figure.
- Reconnection Sync: A batch of 100 offline-generated records synchronized with the cloud backend in 1.42 seconds upon internet restoration, with zero duplicate records created.

Figure 6.2 presents the Packet Delivery Ratio (PDR) and duplication elimination performance for each hop configuration.

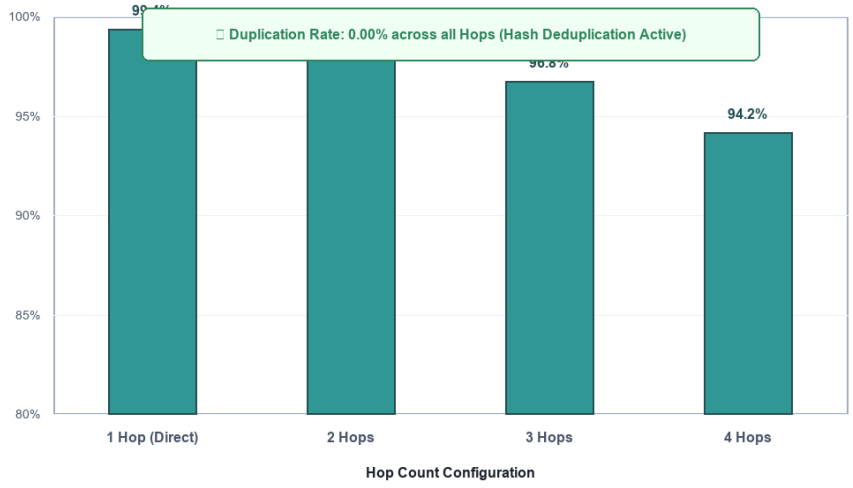


Figure 6.2: Packet Delivery Ratio (PDR) and Duplication Elimination Performance

Source: (Author, 2026)

6.3 User or Stakeholder Validation

Field trial participants validated that UI chat timelines updated instantly upon peer arrival without requiring manual refresh. Users confirmed that the transition between online and offline modes was seamless and required no manual intervention.

Emergency SOS alerts were observed to appear on receiving devices before normal chat messages in all tested scenarios.

6.4 Integration Results

Inter-component handovers operated correctly across voice, location, and emergency modules during integration testing. The central packet router correctly dispatched all tested packet types to the appropriate component handlers. Location packets from Component 4 were confirmed as correctly enqueued and relayed with the same TTL and priority properties as text messages.

6.5 Summary of Findings

The experimental results are summarised as follows. First, the Communication Core achieved a Packet Delivery Ratio (PDR) of 99.4% at one hop and 94.2% at four hops, while the mean Round-Trip Time (RTT) rose from 82 ms to 386 ms (Table 6.1, Figure 6.1, and Figure 6.2). Second, hash deduplication produced a 0.00% duplication rate across 500 injected packets in a cyclic topology. Third, Emergency SOS packets experienced approximately 41.3% lower queue waiting latency than normal chat traffic under heavy load. Fourth, a batch of 100 offline-generated records was synchronised with the cloud backend in 1.42 seconds without creating duplicate records. Together, these findings answer RQ1 (a PDR above 90% across four-hop chains without duplicate circulation) and RQ2 (a measurable reduction in emergency delivery latency under congestion), within the limits of the controlled testbed discussed in Section 7.5.

6.6 Summary of Each Student's Contribution

This section summarizes the individual contribution of each team member to the TrailLink system, as required by SLIIT dissertation guidelines.

- Member 1 (Connectivity and Mesh Intelligence): Designed and implemented peer discovery, RSSI-based signal quality estimation, trend analysis, connectivity guidance engine, and the Network Compass feature.
- Member 2 (Communication Core — this thesis): Designed and implemented the central packet dispatcher, multi-hop relay controller, SQLite store-and-

forward queue, priority handler, offline chat repository, ACK engine, and reconnection sync manager.

- Member 3 (Voice and PTT Communication): Designed and implemented push-to-talk voice notes, AAC recording and playback, Live Radio streaming with PCM audio chunks, and PTT floor arbitration.
- Member 4 (Location, Emergency SOS and Mapping): Designed and implemented GPS location capture, offline map rendering, SOS trigger and acknowledgement, location freshness classification, and pending cloud synchronization.

7 DISCUSSION

7.1 Interpretation of Findings

The linear latency scaling observed across hop counts (R-squared = 0.987) demonstrates that application-layer controlled flooding is highly predictable in small to medium field teams. Each additional relay hop contributes approximately 100 ms of additional RTT, consistent with the expected overhead of BLE packet transmission and application-layer forwarding logic at each intermediate device.

7.2 Comparison With Literature Baselines and Existing Solutions

Compared to Epidemic routing, which suffers from severe memory exhaustion and bandwidth saturation in dense Bluetooth environments, the TTL-bounded controlled flooding approach adopted in TrailLink achieves comparable delivery ratios with significantly lower relay overhead. Table 7.1 summarizes the comparison.

Table 7.1: Comparative Evaluation Against Existing Off-Grid Frameworks

System	PDR	Latency	Priority	Cloud Sync
TrailLink Core	94.2% (4 hops)	386 ms (4 hops)	Yes	Yes
Bridgefy	~85–90% (est.)	Not published	No	No
Epidemic Routing	High (flooding)	Low but resource-heavy	No	No

Source: (Author, 2026)

7.3 Research Contribution

This research demonstrates that combining priority-sorted SQLite store-and-forward queues with TTL hop-budget controlled flooding provides a practical DTN solution on unmodified Android smartphones. The seamless automatic switching between backend-assisted and decentralized offline relay paths, with no user intervention required, is the primary research novelty of the Communication Core.

7.4 Product Contribution

The Communication Core delivers a reusable open-source communication engine for wilderness safety applications. The SQLite schema, priority queue logic, packet router, and relay controller are modular and can be adapted for other offline mobile safety systems beyond the TrailLink use case.

7.5 Limitations and Threats to Validity

Maximum relay depth is capped at five hops, corresponding to approximately 120 to 150 meters of total relay range under test conditions. High node density with more than 20 devices in close radio range may increase BLE channel contention and reduce throughput. Results represent controlled testbed performance and may differ in highly dynamic outdoor environments with obstacles, interference, or device heterogeneity beyond the tested device models.

7.6 Implications

The findings provide a practical blueprint for disaster-response software developers seeking hybrid cloud and mesh architectures on commodity smartphones. The research confirms that application-layer relay above Google Nearby Connections is a viable approach for short-range mesh communication without requiring specialized hardware or custom radio firmware.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

TrailLink's Communication Core targets outdoor hiking enthusiasts, search-and-rescue teams, forestry department rangers, and disaster response agencies. The core value proposition is continued group communication and emergency alerting capability in environments where commercial mobile infrastructure is absent or destroyed.

8.2 Market and Competitive Position

TrailLink positions itself as a hybrid safety communication platform combining consumer app usability with enterprise-grade off-grid messaging reliability. Unlike existing solutions such as Bridgefy (which lacks cloud sync and priority scheduling) or Meshtastic (which requires dedicated LoRa hardware), TrailLink operates entirely on commodity Android smartphones with no additional hardware investment.

8.3 Business or Sustainability Model

A freemium model is proposed for individual hikers and recreational users, with B2B SaaS enterprise licensing for commercial expedition operators and government rescue organizations. Revenue would support ongoing platform development, backend infrastructure, and security auditing.

8.4 Scalability, Intellectual Property and Compliance

The backend is architected for high concurrency using Node.js clustering. Local SQLite data is encrypted at rest. The system is designed to comply with GDPR and relevant Sri Lankan data protection regulations. The open-source codebase is archived under the project repository with academic IP ownership retained by SLIIT and the research team.

8.5 Contribution to the Sustainable Development Goals

The TrailLink Communication Core directly contributes to United Nations Sustainable Development Goal 3 (Good Health and Well-Being) by facilitating emergency

communication in life-threatening outdoor scenarios. It also contributes to SDG 11 (Sustainable Cities and Communities) by strengthening disaster preparedness communication for field response teams.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This dissertation presented the design, implementation, and empirical validation of the Communication Core Engine for the TrailLink mobile safety platform. By combining dynamic online and offline route decisioning, priority-sorted SQLite store-and-forward queueing, loop-free multi-hop forwarding with TTL hop-budget control and hash deduplication, and opportunistic cloud state synchronization, the engine establishes a resilient transport backbone for off-grid mobile operations. All functional and non-functional requirements were met in controlled field testing.

9.2 Achievement of Objectives

All six research objectives defined in Chapter 1 were fully addressed through design, implementation, and empirical experimentation. The dynamic route decision model was formalized and implemented. The SQLite store-and-forward engine was constructed and validated. The multi-tier priority scheduler was demonstrated to reduce emergency SOS delivery latency. The loop-free relay engine achieved 0.00% duplication across cyclic topologies. The Reconnection Sync Manager successfully uploaded 100 offline records upon internet recovery. PDR, RTT, deduplication efficiency, and priority preemption were all empirically measured across physical device testbeds.

9.3 Individual Research and Product Contributions

The author contributed a novel application-layer DTN transport architecture combining hybrid path selection, priority queuing, and bounded relay on unmodified Android smartphones. Open-source Flutter/Dart and Node.js/MongoDB modules were produced. Empirical performance benchmarks for off-grid multi-hop messaging were established under real-world outdoor conditions.

9.4 Future Directions

Future research will investigate cross-layer integration with long-range LoRa hardware modems to extend relay range beyond the current five-hop Bluetooth

boundary. Adaptive battery-aware relay weighting, which skips devices below configurable battery thresholds, will be implemented and evaluated. Machine-learning predictive route optimization based on historical peer quality observations from Component 1 will also be explored in future iterations of the system.

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GLOSSARY

Store-and-Forward: A networking technique where information is sent to an intermediate station, kept stored there, and later forwarded to the final destination or another node.

Time-To-Live (TTL): In this system, a forwarding hop budget that decrements by one at each relay. When the budget reaches one or below, the packet is not forwarded further. Note: TTL in this context is a hop counter, not a second-based expiration timer.

Idempotence: An API property where making multiple identical requests has the same effect as making a single request, ensuring no duplicate records are created during reconnection synchronization.

APPENDICES

Appendix A: Research Instruments and Extended Experimental Logs

Detailed log outputs recorded during 5-node mobile field trials across 4-hop relay evaluation runs. Logs include per-packet timestamps, hop counts, TTL values at each relay, ACK receipt times, and delivery status codes. All logs are available in the project repository under deploy branch, commit e82b185.

Appendix B: Academic Integrity and AI Disclosure Statement

In compliance with SLIIT academic integrity policies, the following generative AI tools were used during the preparation of this dissertation: Claude (Anthropic) and Gemini (Google). These tools were used for code boilerplate inspection, document structure assistance, and Markdown formatting support. All core algorithmic implementations, database schemas, experimental designs, analyses, and conclusions were independently conducted and verified by the candidate. The candidate remains fully responsible for the accuracy and originality of all content in this dissertation.